

Joint Score Structure under OU Diffusion: A Foundational Walk-through of the Gaussian and Laplace $K = 1$ Cases

Session summary — Score-Based Diffusion for Dynamic Objects
Giovanni Mantovani, MSc Thesis, Bocconi University

Abstract

This note records, in a self-contained and pedagogical form, the foundational layer of the thesis on score-based diffusion for dynamic sequences. The aim is to disentangle the role of *Gaussianity* and *Markovianity* in shaping the joint score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. We treat in full the Gaussian AR(1) sequence as a benchmark, and then carry out the Laplace warm-up at $K = 1$, where the noised distribution becomes genuinely non-Gaussian and the score acquires nonlinearity. The Tweedie identity is derived in general and used to organise both cases. The analysis of $K \geq 2$ for the Laplace model — where Markovianity proper enters the picture — is left to a subsequent session.

Contents

1	Setup and notation	1
2	The Gaussian AR(1) benchmark	2
2.1	The clean distribution	2
2.2	Noised distribution and joint score	2
2.3	Spectral analysis	3
2.4	Sparsity, density and the locality verdict	3
3	The Laplace warm-up at $K = 1$	3
3.1	Why $K = 1$?	3
3.2	Setup	4
3.3	Closed form via a truncated-Gaussian decomposition	4
3.4	Sanity check via the integral representation	4
3.5	The score: Tweedie identity	5
3.6	What the Laplace $K = 1$ case teaches	5
4	Outlook: towards $K \geq 2$	5
A	Why $\partial_x \log p(x a) = -(x - \mu a)/\sigma^2$	6
B	Mills-ratio asymptotic of Φ	6

1 Setup and notation

We study a sequence $\mathbf{a} = (a_0, a_1, \dots, a_{K-1}) \in \mathbb{R}^K$ drawn from a clean distribution P_0 . The notation deliberately distinguishes the *clean* variable $\mathbf{a}$ from the *noised* variable $\mathbf{x}$, so that all derivations remain unambiguous when both appear in the same expression.

Forward process. Throughout, the corruption is the variance-preserving Ornstein–Uhlenbeck (OU) SDE

$$d\mathbf{x}_t = -\mathbf{x}_t dt + \sqrt{2} d\mathbf{W}_t, \quad \mathbf{x}_0 = \mathbf{a}, \quad (1)$$

which admits the closed-form solution

$$\mathbf{x}_t = e^{-t} \mathbf{a} + \sqrt{1 - e^{-2t}} \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, I), \quad (2)$$

with $\boldsymbol{\xi}$ independent of $\mathbf{a}$. We adopt the shorthand

$$\mu := e^{-t}, \quad \sigma^2 := 1 - e^{-2t}, \quad (3)$$

so that the conditional law is $P(\mathbf{x} \mid \mathbf{a}) = \mathcal{N}(\mu \mathbf{a}, \sigma^2 I)$.

Object of interest. The thesis studies the *joint* score

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) \in \mathbb{R}^K, \quad (4)$$

not the per-frame marginal score $\nabla_{x_u} \log p_t(x_u)$. The component S_u is the partial derivative of $\log P_t$ with respect to x_u *holding all other frames fixed*; it is precisely this conditioning that carries the inter-frame structure inherited from P_0 .

2 The Gaussian AR(1) benchmark

2.1 The clean distribution

The clean process is the stationary AR(1) recursion

$$a_u = \alpha a_{u-1} + \varepsilon_u, \quad \varepsilon_u \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad |\alpha| < 1, \quad (5)$$

with $a_0 \sim \mathcal{N}(0, \frac{1}{1-\alpha^2})$ chosen to enforce stationarity: $\text{Var}(a_u) = \alpha^2 \text{Var}(a_{u-1}) + 1$ forces $\sigma_{\text{stat}}^2 = \frac{1}{1-\alpha^2}$.

Joint distribution. Since a_0 is Gaussian and each step in (5) is a linear combination of independent Gaussians, $P_0(\mathbf{a})$ is a multivariate Gaussian. By induction $\mathbb{E}[a_u] = 0$ for all u , hence the joint law is determined by the covariance matrix alone:

$$P_0(\mathbf{a}) = \mathcal{N}(\mathbf{0}, \Sigma_0), \quad (\Sigma_0)_{uu'} = \frac{\alpha^{|u-u'|}}{1-\alpha^2}. \quad (6)$$

The covariance is computed by unrolling the recursion: for $u \geq u'$,

$$a_u = \alpha^{u-u'} a_{u'} + \sum_{j=0}^{u-u'-1} \alpha^j \varepsilon_{u-j},$$

and the innovations beyond u' are independent of $a_{u'}$, leaving $\mathbb{E}[a_u a_{u'}] = \alpha^{u-u'} / (1 - \alpha^2)$.

Markov structure. The matrix Σ_0 is *dense*: every pair of frames is correlated. The Markov property is encoded not in Σ_0 but in its inverse $J_0 = \Sigma_0^{-1}$, the *precision matrix*. This is the matrix of *conditional* dependencies: starting from the Gaussian density $P_0(\mathbf{a}) \propto \exp(-\frac{1}{2} \mathbf{a}^\top J_0 \mathbf{a})$, completing the square in a_u gives

$$\mathbb{E}[a_u \mid \mathbf{a}_{-u}] = -\frac{1}{(J_0)_{uu}} \sum_{u' \neq u} (J_0)_{uu'} a_{u'}, \quad \text{Var}(a_u \mid \mathbf{a}_{-u}) = \frac{1}{(J_0)_{uu}}, \quad (7)$$

so $(J_0)_{uu'} = 0$ if and only if $a_u \perp a_{u'} \mid \mathbf{a}_{-\{u, u'\}}$. The Markov property of (5) predicts $(J_0)_{uu'} = 0$ for $|u - u'| > 1$, i.e. J_0 tridiagonal.

Explicit form of J_0 . Using the Markov factorisation $P_0(\mathbf{a}) = p(a_0) \prod_{u=1}^{K-1} p(a_u | a_{u-1})$ and reading off coefficients,

$$J_0 = \Sigma_0^{-1} = \begin{pmatrix} 1 & -\alpha & 0 & \cdots & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha & \cdots & 0 \\ 0 & -\alpha & 1 + \alpha^2 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & \cdots & -\alpha & 1 \end{pmatrix}, \quad (8)$$

which is indeed tridiagonal, with $(1 + \alpha^2)$ on the bulk diagonal, 1 at the boundary, and $-\alpha$ on the off-diagonals.

2.2 Noised distribution and joint score

Since (2) is a linear combination of two independent Gaussian vectors, $P_t(\mathbf{x})$ is again Gaussian. Independence of $\mathbf{a}$ and $\boldsymbol{\xi}$ gives

$$\Sigma_t = \mathbb{E}[\mathbf{x}_t \mathbf{x}_t^\top] = e^{-2t} \Sigma_0 + (1 - e^{-2t}) I, \quad P_t(\mathbf{x}) = \mathcal{N}(\mathbf{0}, \Sigma_t). \quad (9)$$

Theorem 1 (Gaussian AR(1) joint score). *For the Gaussian AR(1) clean distribution corrupted by the OU process,*

$$\boxed{S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -\Sigma_t^{-1} \mathbf{x}} \quad (10)$$

which is linear in $\mathbf{x}$ for every $t \geq 0$.

2.3 Spectral analysis

Let $\Sigma_0 = Q \Lambda Q^\top$ be the eigendecomposition with Q orthogonal and $\Lambda = \text{diag}(\lambda_k)$. Writing $I = Q Q^\top$,

$$\Sigma_t = Q(e^{-2t} \Lambda + \sigma_t^2 I) Q^\top, \quad \Sigma_t^{-1} = Q \text{diag}\left(\frac{1}{e^{-2t} \lambda_k + \sigma_t^2}\right) Q^\top. \quad (11)$$

In the eigenbasis $\tilde{\mathbf{x}} = Q^\top \mathbf{x}$ the score decouples completely into K independent scalar denoisers

$$\tilde{S}_k(\tilde{\mathbf{x}}, t) = -\frac{\tilde{x}_k}{e^{-2t} \lambda_k + \sigma_t^2}. \quad (12)$$

The coupling visible in coordinate space is generated entirely by the two rotations $Q^\top$ and Q that surround the diagonal core; the underlying operation is trivially a per-mode rescaling.

2.4 Sparsity, density and the locality verdict

Object	At $t = 0$	At $t > 0$
Σ_t	dense	dense
Σ_t^{-1}	tridiagonal	dense (generically)
$S(\mathbf{x}, t) = -\Sigma_t^{-1} \mathbf{x}$	local	non-local

The only sparse object in the entire Gaussian story is J_0 . The addition of the isotropic perturbation $\sigma_t^2 I$ to Σ_0 preserves the eigenvectors but shifts eigenvalues uniformly; under inversion the eigenvector matrix Q — generically dense — *recouples* all coordinates. Sparsity is destroyed instantly for any $t > 0$.

Remark (Disentangling Gaussianity from Markovianity). The Gaussian benchmark conflates two structural properties that are logically distinct:

- *Gaussianity* is responsible for the linearity of the score $S(\mathbf{x}, t) = -\Sigma_t^{-1} \mathbf{x}$ at *all* times. It has nothing to do with the Markov structure of P_0 .

- *Markovianity* is responsible for the tridiagonal sparsity of J_0 at $t = 0$. It has nothing to do with the score being linear.

Locality is therefore a property of the clean-time score only, and it is a manifestation of Markovianity, not of Gaussianity. To probe Markovianity in isolation we must break the Gaussian assumption.

3 The Laplace warm-up at $K = 1$

3.1 Why $K = 1$?

For $K = 1$ there is no sequence and no Markov structure. The model is pedagogically essential nonetheless: it is the smallest case in which the noised distribution is genuinely non-Gaussian and the score becomes nonlinear, while remaining exactly tractable. It is the *atomic building block* from which $K \geq 2$ will be assembled, since the Laplace AR(1) factorises into a Laplace prior on a_0 and shifted-Laplace conditionals $p(a_u | a_{u-1})$.

3.2 Setup

The clean variable is

$$a \sim \text{Laplace}(0, 1), \quad p_0(a) = \frac{1}{2}e^{-|a|}, \quad (13)$$

and the OU forward step yields

$$p(x | a) = \mathcal{N}(\mu a, \sigma^2), \quad p_t(x) = \int_{-\infty}^{+\infty} p(x | a) p_0(a) da. \quad (14)$$

3.3 Closed form via a truncated-Gaussian decomposition

The integrand is the product of a Gaussian in a and the kink $e^{-|a|}$. The standard procedure is:

1. Split at $a = 0$: $|a| = a$ on $\mathbb{R}_+$, $|a| = -a$ on $\mathbb{R}_-$.
2. On each half-line, complete the square in a . The exponent becomes a centred Gaussian shape times an a -independent constant.
3. Recognise the resulting integrals as truncated Gaussians and express them through the standard normal CDF $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$.

Completing the square (positive branch). For $a > 0$,

$$-\frac{(x-\mu a)^2}{2\sigma^2} - a = -\frac{\mu^2}{2\sigma^2}(a - m_+)^2 + C_+,$$

where, expanding and matching a^2 and a coefficients,

$$m_+ = \frac{x\mu - \sigma^2}{\mu^2}, \quad C_+ = -\frac{x}{\mu} + \frac{\sigma^2}{2\mu^2}. \quad (15)$$

The negative branch yields $m_- = (x\mu + \sigma^2)/\mu^2$ and $C_- = +x/\mu + \sigma^2/(2\mu^2)$ by parity.

Truncated Gaussian formula. For mean m , variance v and threshold c ,

$$\int_c^{+\infty} e^{-\frac{(a-m)^2}{2v}} da = \sqrt{2\pi v} \Phi\left(\frac{m-c}{\sqrt{v}}\right). \quad (16)$$

With $v = \sigma^2/\mu^2$ and $c = 0$ this gives $\sqrt{2\pi} \sigma/\mu \cdot \Phi(m_+ \mu/\sigma)$ for the positive branch and $\sqrt{2\pi} \sigma/\mu \cdot \Phi(-m_- \mu/\sigma)$ for the negative one. Assembling and simplifying the prefactor $\frac{1}{2\sqrt{2\pi}\sigma} \cdot \frac{\sqrt{2\pi}\sigma}{\mu} = \frac{1}{2\mu}$,

$$\boxed{p_t(x) = \frac{1}{2\mu} \left[e^{C_+} \Phi\left(\frac{m_+\mu}{\sigma}\right) + e^{C_-} \Phi\left(-\frac{m_-\mu}{\sigma}\right) \right]} \quad (17)$$

3.4 Sanity check via the integral representation

Limit analysis is cleanest on the original convolution (14), not on the closed form, where competing divergences obscure the physics.

$t \rightarrow 0$. As $\sigma \rightarrow 0$ the Gaussian kernel $\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu a)^2/(2\sigma^2)}$ is a sequence of increasingly narrow bumps of unit area centred at $a = x/\mu \rightarrow x$, hence converges (in distribution) to the Dirac mass $\delta(a - x)$. The sifting property gives $p_0(x) = \frac{1}{2}e^{-|x|}$, recovering the Laplace prior.

$t \rightarrow \infty$. As $\mu \rightarrow 0$ the term μa inside the Gaussian vanishes uniformly in a , so the kernel becomes a -independent, $\frac{1}{\sqrt{2\pi}\sigma}e^{-x^2/(2\sigma^2)}$, and factors out of the integral, leaving only the Laplace normalisation $\int e^{-|a|}da = 2$. Hence $p_\infty(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, the standard Gaussian attractor of the OU process.

3.5 The score: Tweedie identity

Rather than differentiate the closed form directly, one differentiates the convolution (14). Using $\partial_x p(x | a) = -\frac{x-\mu a}{\sigma^2} p(x | a)$ and Bayes' theorem $p(a | x) = p(x | a)p_0(a)/p_t(x)$,

$$S(x, t) = \frac{\partial_x p_t(x)}{p_t(x)} = -\frac{1}{\sigma^2} [x - \mu \mathbb{E}[a | x]]. \quad (18)$$

Theorem 2 (Tweedie identity). *For any prior p_0 and the OU forward channel $x = \mu a + \sigma \xi$ with $\xi \sim \mathcal{N}(0, 1)$,*

$$\boxed{S(x, t) = \nabla_x \log p_t(x) = -\frac{x - \mu \mathbb{E}[a | x]}{\sigma^2} = -\frac{\mathbb{E}[\xi | x]}{\sigma}} \quad (19)$$

The two equivalent forms are connected by the algebraic identity $x - \mu a = \sigma \xi$ and the linearity of conditional expectation. The interpretation is striking: *the score is the posterior expected noise, normalised by the noise scale.*

3.6 What the Laplace $K = 1$ case teaches

- The same Tweedie identity governs both Gaussian and Laplace priors. The entire dependence on p_0 is concentrated in the posterior mean $\mathbb{E}[a | x]$.
- For Gaussian p_0 , the posterior mean is affine in x , so the score is linear. For Laplace p_0 the posterior mean is *nonlinear* in x : a smooth interpolation between $-\text{sign}(x)$ at $t = 0$ (Laplace score) and $-x$ at $t = \infty$ (Gaussian score).
- Thus, breaking Gaussianity *breaks linearity*, even though the Tweedie skeleton survives. The thesis question — whether Markovianity leaves a structural imprint — requires us to push to $K \geq 2$, where coupling between frames can finally be probed.

4 Outlook: towards $K \geq 2$

The Tweedie identity extends without change to vectors:

$$S(\mathbf{x}, t) = -\frac{\mathbf{x} - \mu \mathbb{E}[\mathbf{a} | \mathbf{x}]}{\sigma^2}. \quad (20)$$

At $K = 2$ the prior factorises by the Markov property as

$$P_0(a_0, a_1) = p_0(a_0) p(a_1 | a_0) = \frac{1}{2}e^{-|a_0|} \cdot \frac{1}{2}e^{-|a_1 - \alpha a_0|}, \quad (21)$$

i.e. a Laplace prior on a_0 together with a shifted-Laplace conditional. The vector posterior mean $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] \in \mathbb{R}^2$ is the object where Markov structure can finally make itself felt: each component will depend on *both* x_0 and x_1 , and the precise way in which this coupling enters is what the next session has to study.

Working hypotheses to verify next.

- Whether the joint Hessian $H_t(\mathbf{x}) = -\nabla^2 \log P_t(\mathbf{x})$ inherits any remnant of the tridiagonal structure of the Gaussian precision matrix J_0 — now in a position-dependent, non-affine form.
- Whether nonlinearity is concentrated in the innovation/residual channel $a_1 - \alpha a_0$, while the marginal of a_0 contributes a more standard piece.
- Whether the K=2 obstruction reported in earlier project work (a non-vanishing off-diagonal block under symbolic Schur reduction) can be interpreted as Markovianity *deforming*, rather than destroying, locality.

A Why $\partial_x \log p(x \mid a) = -(x - \mu a)/\sigma^2$

For the Gaussian conditional $p(x \mid a) = \mathcal{N}(\mu a, \sigma^2)$, $\log p(x \mid a) = -\frac{(x - \mu a)^2}{2\sigma^2} + \text{const}$, so $\partial_x \log p(x \mid a) = -(x - \mu a)/\sigma^2$. The log-derivative trick $\partial_x p = p \partial_x \log p$ then converts the differentiation under the integral sign into an expectation against the posterior, giving Tweedie.

B Mills-ratio asymptotic of Φ

For $z \rightarrow -\infty$, integration by parts on $\int_{|z|}^{\infty} e^{-u^2/2} du$ with $f = 1/u$, $dg = u e^{-u^2/2} du$ produces $\int_{|z|}^{\infty} e^{-u^2/2} du = e^{-z^2/2}/|z| + O(e^{-z^2/2}/|z|^3)$, hence $\Phi(z) \sim \frac{1}{\sqrt{2\pi}|z|} e^{-z^2/2}$. This is the precise rate at which the Gaussian CDF vanishes in its left tail and is the engine behind the cancellation of the apparent $0 \cdot \infty$ blow-ups in the closed-form $p_t(x)$ as $t \rightarrow \infty$.